Eulerian Coordinate Derivation and Sub–Shock Patching Method

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1 Problem Setup and Self-Similar Ansatz

We consider the one-temperature radiation-hydrodynamics equations in the diffusion approximation, written in Lagrangian (mass) coordinates (m, t):

$$\frac{\partial v}{\partial t} - \frac{\partial u}{\partial m} = 0 \tag{1}$$

$$\frac{\partial u}{\partial t} + \frac{\partial p}{\partial m} = 0 \tag{2}$$

$$\frac{\partial e}{\partial t} + p\,\frac{\partial v}{\partial t} = -\frac{\partial s}{\partial m} \tag{3}$$

where $v = 1/\rho$ is the specific volume, u is the velocity, p the pressure, and e the specific internal energy. The radiation energy flux is:

$$s(m,t) = -\frac{c_{\text{light}}}{3\kappa_R}\,\frac{\partial}{\partial m}\left(a_R\,T^4\right). \tag{4}$$

The material model consists of an ideal gas EOS with adiabatic index γ (writing $r = \gamma - 1$), power-law Rosseland opacity, and power-law specific internal energy:

$$p\,v = r\,e, \tag{5}$$

$$\frac{1}{\kappa_R(T,\rho)} = g\,T^{\alpha}\,\rho^{-\lambda} = g\,T^{\alpha}\,v^{\lambda}, \tag{6}$$

$$e(T,\rho) = f\,T^{\beta}\,\rho^{-\mu} = f\,T^{\beta}\,v^{\mu}. \tag{7}$$

Defining the derived constants:

$$n = \frac{4+\alpha}{\beta}, \quad k = 1-\mu, \quad q = k\,n + \lambda - 1, \tag{8}$$

the radiation flux can be rewritten as:

$$s(m,t) = -B\,v^{\lambda}\left(p\,v^{k}\right)^{n-1}\frac{\partial}{\partial m}\left(p\,v^{k}\right), \tag{9}$$

where the diffusion prefactor is:

$$B = \frac{16\,\sigma_{\text{sb}}\,g}{3\,\beta\,(r\,f)^{n}}. \tag{10}$$

The boundary conditions are:

$$T(0,t) = T_0\,t^{\tau}, \qquad p(0,t) = 0. \tag{11}$$

1.1 Dimensional Constants and Self-Similar Coordinate

Two dimensional constants characterise the problem:

$$A = T_0\,(r\,f)^{1/\beta}, \qquad B = \frac{16\,\sigma_{\mathrm{sb}}\,g}{3\,\beta\,(r\,f)^n}. \tag{12}$$

The self-similar coordinate is:

$$\xi = m\,A^a\,B^b\,t^c, \tag{13}$$

where the exponents (a, b, c) are determined by dimensional analysis:

$$a = \frac{2\beta - \beta\lambda - 8 - 2\alpha + \mu(4+\alpha)}{4 + 2\lambda - 4\mu}, \tag{14}$$

$$b = \frac{\mu - 2}{4 + 2\lambda - 4\mu}, \tag{15}$$

$$c = \frac{3\mu - 2\lambda - 2 + \tau\big(2(\beta - \alpha - 4) - \beta\lambda + \mu(4+\alpha)\big)}{4 + 2\lambda - 4\mu}. \tag{16}$$

The self-similar profiles are:

$$v(m,t) = V(\xi)\,A^{a_1}\,B^{b_1}\,t^{c_1}, \tag{17}$$

$$u(m,t) = U(\xi)\,A^{a_2}\,B^{b_2}\,t^{c_2}, \tag{18}$$

$$p(m,t) = P(\xi)\,A^{a_3}\,B^{b_3}\,t^{c_3}, \tag{19}$$

with the exponent relations (from the mass, momentum, and energy equations):

$$a_2 = a_1 - a, \qquad b_2 = b_1 - b, \qquad c_2 + 1 = c_1 - c, \tag{20}$$

$$a_3 = a_1 - 2a, \qquad b_3 = b_1 - 2b, \qquad c_3 + 2 = c_1 - 2c. \tag{21}$$

2 Relation to Eulerian Coordinate – Positions in Space

2.1 Ablated Mass

The ablated (heat-front) mass is:

$$\xi_f = m_f(t)\,A^a\,B^b\,t^c, \qquad \Longrightarrow \qquad m_f(t) = A^{-a}\,B^{-b}\,t^{-c}\,\xi_f. \tag{22}$$

For the heat wave to propagate outwards we require $-c > 0$, i.e. $c < 0$.

2.2 Exact Position for a Given Mass

The Eulerian position of a Lagrangian element m relative to the boundary is obtained by integrating the specific volume:

$$x(t,m) - x_b(t) = \int_0^m v(m',t)\,dm'. \tag{23}$$

Substituting the self-similar profile $v(m,t) = V(\xi)\,A^{a_1}B^{b_1}t^{c_1}$ and changing variables from m' to ξ' using $m' = A^{-a}B^{-b}t^{-c}\,\xi'$, hence $dm' = A^{-a}B^{-b}t^{-c}\,d\xi'$:

$$\begin{aligned} x(t,m) - x_b(t) &= \int_0^{\xi(m)} V(\xi')\,A^{a_1}\,B^{b_1}\,t^{c_1}\,A^{-a}\,B^{-b}\,t^{-c}\,d\xi' \\ &= A^{a_1-a}\,B^{b_1-b}\,t^{c_1-c}\int_0^{\xi(m)} V(\xi')\,d\xi'. \end{aligned} \tag{24}$$

Using the power relations (20), this simplifies to:

$$\boxed{x(t,m) - x_b(t) = A^{a_2}\,B^{b_2}\,t^{c_2+1}\int_0^{\xi(m)} V(\xi')\,d\xi'.} \tag{25}$$

2.3 Distance from Boundary to Ablation Front

Setting $m = m_f(t)$ in (25):

$$\Delta x_f(t) = x(t,m_f) - x_b(t) = A^{a_2}\,B^{b_2}\,t^{c_2+1}\int_0^{\xi_f} V(\xi')\,d\xi'. \tag{26}$$

2.4 Boundary Position

The boundary velocity is:

$$u_b(t) \equiv u(m{=}0,t) = U(0)\,A^{a_2}\,B^{b_2}\,t^{c_2}. \tag{27}$$

Integrating in time:

$$x_b(t) = \int_0^t u_b(t')\,dt' = U(0)\,A^{a_2}\,B^{b_2}\int_0^t t'^{\,c_2}\,dt' = \frac{A^{a_2}\,B^{b_2}}{c_2+1}\,U(0)\,t^{c_2+1}. \tag{28}$$

This requires $c_2 > -1$ for convergence.

2.5 Ablation Front Position

We expect the ablation front to remain at $x = 0$:

$$x_f(t) \equiv x_b(t) + \Delta x_f(t) = 0. \tag{29}$$

Combining (28) and (26):

$$x_f(t) = A^{a_2}\, B^{b_2}\, t^{c_2+1} \left[\frac{U(0)}{c_2+1} + \int_0^{\xi_f} V(\xi')\, d\xi' \right] = 0. \tag{30}$$

This is satisfied by the integral identity (proven below):

$$\boxed{\int_0^{\xi_f} V(\xi')\, d\xi' = -\frac{U(0)}{c_2+1}.} \tag{31}$$

Since $U(0) < 0$ (boundary moves in the negative direction) and $c_2 + 1 > 0$, the integral is positive as expected.

2.6 Ablation Front Velocity

Since $x_f(t) \equiv 0$ for all t:

$$v_f(t) = \frac{dx_f}{dt} = 0. \tag{32}$$

2.7 Positions as a Function of Time (Final Result)

For any Lagrangian element with mass $m < m_f(t)$ (inside the ablated region):

$$x(t,m) = A^{a_2}\, B^{b_2}\, t^{c_2+1} \left[\frac{U(0)}{c_2+1} + \int_0^{\xi(m)} V(\xi')\, d\xi' \right], \tag{33}$$

where $\xi(m) = m\, A^a\, B^b\, t^c$.

Using the integral identity (derived in §3):

$$\int_0^{\xi_0} V(\xi)\, d\xi = \frac{1}{c_1 - c} \Big[U(\xi_0) - U(0) - c\, \xi_0\, V(\xi_0) \Big], \tag{34}$$

and substituting $c_1 - c = c_2 + 1$, we obtain:

$$x(t,m) = A^{a_2}\, B^{b_2}\, t^{c_2+1} \left[\frac{U(0)}{c_2+1} + \frac{U(\xi(m)) - U(0) - c\, \xi(m)\, V(\xi(m))}{c_2+1} \right], \tag{35}$$

which simplifies to the closed-form expression:

$$\boxed{x(t,m) = \frac{A^{a_2}\,B^{b_2}\,t^{c_2+1}}{c_2+1}\Big[U\big(\xi(m)\big) - c\,\xi(m)\,V\big(\xi(m)\big)\Big]} \tag{36}$$

for $m \le m_f(t)$, and $x(t,m) = 0$ for $m \ge m_f(t)$.

Numerical caveat

At the ablation front, $U_f = V_f = 0$ exactly, so $x(t, m_f) = 0$. However, due to numerical errors, U_f and V_f may not be exactly zero, and the factor $U(\xi) - c\,\xi\,V(\xi)$ can be slightly positive near $\xi \approx \xi_f$ (since $U \le 0$, $V \ge 0$, but $c < 0$). In practice, if this factor is slightly positive, it should be clamped to zero to ensure $x(t, m \ge m_f) = 0$.

3 Calculation of the $V(\xi)$ Integral

From the self-similar mass (continuity) equation:

$$c\,\xi\,V' + c_1\,V - U' = 0. \tag{37}$$

We can write:

$$V(\xi) = \frac{1}{c_1}\big(U' - c\,\xi\,V'\big) = p_\xi\,\xi\,V' + q_\xi\,U', \tag{38}$$

with $p_\xi = -c/c_1$ and $q_\xi = 1/c_1$.

Integrating from 0 to ξ_0:

$$\int_0^{\xi_0} V(\xi)\,d\xi = p_\xi \int_0^{\xi_0} \xi\,V'(\xi)\,d\xi + q_\xi\big[U(\xi_0) - U(0)\big]. \tag{39}$$

Integration by parts for the first term:

$$\int_0^{\xi_0} \xi\,V'(\xi)\,d\xi = \big[\xi\,V(\xi)\big]_0^{\xi_0} - \int_0^{\xi_0} V(\xi)\,d\xi. \tag{40}$$

Note that $\lim_{\xi\to 0} \xi\,V(\xi) = 0$, since the integral $\int_0^{\xi_0} V(\xi)\,d\xi$ must be finite (it represents a physical position), requiring $V(\xi) \sim V_0\,\xi^w$ with $w > -1$ near the origin, hence $\xi\,V(\xi) \sim \xi^{w+1} \to 0$.

Collecting terms:

$$(1 + p_\xi) \int_0^{\xi_0} V(\xi)\,d\xi = p_\xi\,\xi_0\,V(\xi_0) + q_\xi\big[U(\xi_0) - U(0)\big]. \tag{41}$$

Since:

$$\frac{p_\xi}{1+p_\xi} = \frac{-c}{c_1 - c}, \qquad \frac{q_\xi}{1+p_\xi} = \frac{1}{c_1 - c}, \tag{42}$$

we obtain:

$$\boxed{\int_0^{\xi_0} V(\xi)\, d\xi = \frac{1}{c_1 - c}\Big[U(\xi_0) - U(0) - c\,\xi_0\, V(\xi_0)\Big].} \tag{43}$$

For $\xi_0 = \xi_f$, using $V_f = U_f = 0$ and $c_1 - c = c_2 + 1$:

$$\int_0^{\xi_f} V(\xi)\, d\xi = -\frac{U(0)}{c_2 + 1} > 0, \tag{44}$$

since $U(0) < 0$ and $c_2 + 1 > 0$.

4 Bath Temperature for the Marshak Boundary Condition

4.1 Derivation

Let the imposed surface temperature drive be:

$$T_s(t) = T_0\, t^\tau. \tag{45}$$

The Marshak boundary condition is:

$$\frac{c}{4}\, F_{\mathrm{inc}}(t) = \frac{a_R\, c}{4}\, T_s^4(t) + \frac{c}{2}\, F_s(t), \tag{46}$$

where $F_s(t)$ is the outgoing radiation flux at the surface. For a bath temperature T_{bath}:

$$F_{\mathrm{inc}}(t) = a_R\, c\, T_{\mathrm{bath}}^4(t). \tag{47}$$

Therefore:

$$T_{\mathrm{bath}}^4(t) = T_s^4(t) + \frac{2\, F_s(t)}{a_R\, c}. \tag{48}$$

Using $a_R\, c = 4\,\sigma_{\mathrm{sb}}$:

$$T_{\mathrm{bath}}^4(t) = T_s^4(t) + \frac{F_s(t)}{2\,\sigma_{\mathrm{sb}}}. \tag{49}$$

4.2 Self-Similar Flux at the Surface

The self-similar radiation energy flux is:

$$S(\xi) = -P^{n-1}\, V^{q}\big(V\, P' + k\, P\, V'\big), \tag{50}$$

and the dimensional flux is:

$$s(m,t) = S(\xi)\, A^{a_S}\, B^{b_S}\, t^{c_S}, \tag{51}$$

with flux powers:

$$a_S = a + (q+1)\, a_1 + n\, a_3, \tag{52}$$
$$b_S = 1 + b + (q+1)\, b_1 + n\, b_3, \tag{53}$$
$$c_S = c + (q+1)\, c_1 + n\, c_3. \tag{54}$$

At the surface/origin:

$$F_s(t) = s(0,t) = S_0\, A^{a_S}\, B^{b_S}\, t^{c_S}, \tag{55}$$

where $S_0 = S(\xi{=}0)$.

4.3 Final Bath Temperature

Substituting into (49):

$$T_{\rm bath}^4(t) = T_0^4\, t^{4\tau} + \frac{S_0\, A^{a_S}\, B^{b_S}}{2\,\sigma_{\rm sb}}\, t^{c_S}. \tag{56}$$

Factoring out $T_0^4\, t^{4\tau}$:

$$T_{\rm bath}^4(t) = T_0^4\, t^{4\tau}\left[1 + \frac{S_0\, A^{a_S}\, B^{b_S}}{2\,\sigma_{\rm sb}\, T_0^4}\, t^{c_S - 4\tau}\right]. \tag{57}$$

Therefore:

$$\boxed{T_{\rm bath}(t) = T_0\, t^{\tau}\left[1 + \mathcal{B}_{\rm bath}\, t^{\eta_{\rm bath}}\right]^{1/4}} \tag{58}$$

with:

$$\mathcal{B}_{\rm bath} = \frac{S_0\, A^{a_S}\, B^{b_S}}{2\,\sigma_{\rm sb}\, T_0^4}, \tag{59}$$

$$\eta_{\rm bath} = c_S - 4\,\tau = \frac{\tau\big[3\beta\lambda + 2\beta + 2\alpha - 8 - 8\lambda + 4\mu - 3\alpha\mu\big] + 3\mu - 2}{4 + 2\lambda - 4\mu}. \tag{60}$$

The explicit form of $\mathcal{B}_{\rm bath}$ is:

$$\mathcal{B}_{\rm bath} = \frac{S_0}{2\,\sigma_{\rm sb}\,T_0^4}\left[T_0\big((\gamma-1)\,f\big)^{1/\beta}\right]^{a_S}\left[\frac{16\,\sigma_{\rm sb}\,g}{3\,\beta\,\big((\gamma-1)\,f\big)^n}\right]^{b_S}. \tag{61}$$

The correction term $\mathcal{B}_{\rm bath}\,t^{\eta_{\rm bath}}$ is dimensionless and measures the ratio of the outgoing surface flux to the blackbody radiation:

$$\mathcal{B}_{\rm bath}\,t^{\eta_{\rm bath}} = \frac{F_s(t)}{2\,\sigma_{\rm sb}\,T_s^4(t)} = \frac{2\,F_s(t)}{a_R\,c\,T_s^4(t)}. \tag{62}$$

5 Self-Similar Radiation Flux

Starting from the dimensional flux:

$$s(m,t) = -B\left(p\,v^k\right)^{n-1} v^\lambda\,\frac{\partial}{\partial m}\left(p\,v^k\right), \tag{63}$$

and substituting the self-similar profiles:

$$v = V\,A^{a_1}\,B^{b_1}\,t^{c_1}, \qquad p = P\,A^{a_3}\,B^{b_3}\,t^{c_3}, \tag{64}$$

$$\frac{\partial v}{\partial m} = A^{a_1+a}\,B^{b_1+b}\,t^{c_1+c}\,V', \qquad \frac{\partial p}{\partial m} = A^{a_3+a}\,B^{b_3+b}\,t^{c_3+c}\,P', \tag{65}$$

we find after collecting powers:

$$s(m,t) = A^{a+(q+1)a_1+n\,a_3}\,B^{1+b+(q+1)b_1+n\,b_3}\,t^{c+(q+1)c_1+n\,c_3}\,S(\xi), \tag{66}$$

where the dimensionless self-similar flux profile is:

$$\boxed{S(\xi) = -P^{n-1}(\xi)\,V^q(\xi)\,\big[V(\xi)\,P'(\xi) + k\,P(\xi)\,V'(\xi)\big].} \tag{67}$$

6 Energy Integrals

The internal energy is:

$$E_{\rm in}(t) = \int_0^{m_f(t)} e(m,t)\,dm = \frac{1}{r}\int_0^{m_f(t)} p\,v\,dm = A^{2a_2-a}\,B^{2b_2-b}\,t^{2c_2-c}\int_0^{\xi_f}\frac{P\,V}{r}\,d\xi. \tag{68}$$

The kinetic energy is:

$$E_k(t) = \int_0^{m_f(t)}\frac{1}{2}\,u^2\,dm = A^{2a_2-a}\,B^{2b_2-b}\,t^{2c_2-c}\int_0^{\xi_f}\frac{U^2}{2}\,d\xi. \tag{69}$$

Both energies share the same temporal power law t^{2c_2-c}, hence the ratio E_k/E_{in} is constant in time.

The total energy integral satisfies the exact identity (for any $\xi_0 \le \xi_f$):

$$\boxed{\int_0^{\xi_0} \left(\frac{PV}{r} + \frac{U^2}{2} \right) d\xi = \frac{1}{2c_2 - c} \left[-S(\xi) - P(\xi)\,U(\xi) - c\,\xi \left(\frac{PV}{r} + \frac{U^2}{2} \right) \right]_0^{\xi_0}.} \tag{70}$$

7 Stitching the Subsonic Heat Wave and the Shock

In the subsonic ablation problem, the heat wave drives material outward (in the negative-x direction in our convention), creating an ablation front at $x_f(t) \equiv 0$. Beyond this front, the unperturbed material is at rest. However, in a realistic problem, the ablated material sweeps up the ambient gas, generating a **piston-driven shock wave** that propagates ahead of the ablation front. The full problem therefore consists of two self-similar regions that must be patched together.

7.1 The Two Regions

1. **Subsonic ablation (heat wave) region** ($m < m_f(t)$):
   Governed by the radiation-hydrodynamics self-similar ODEs derived above, with profiles $V(\xi)$, $U(\xi)$, $P(\xi)$ and the self-similar coordinate $\xi = m\,A^a\,B^b\,t^c$.

2. **Piston shock region** ($m \ge m_f(t)$):
   Governed by the pure-hydrodynamic (adiabatic) piston shock self-similar equations. The piston shock solution uses a different self-similar coordinate

   $$\xi_{\text{shock}} = m\,v_0^{1/(2-\omega)}\,p_0^{(\omega-1)/(2-\omega)}\,t^{(\tau_s+2)(\omega-1)/(2-\omega)}, \tag{71}$$

   with shock profiles $V_{\text{shock}}(\xi_{\text{shock}})$, $U_{\text{shock}}(\xi_{\text{shock}})$, $P_{\text{shock}}(\xi_{\text{shock}})$.

7.2 Matching Conditions at the Ablation Front

The patching is performed at the ablation front mass $m = m_f(t)$. The key matching condition is that the **pressure at the origin of the shock problem** equals the **pressure at the ablation front of the subsonic heat wave**:

$$p_{\text{shock}}(0,t) = P_f\,A^{a_3}\,B^{b_3}\,t^{c_3}, \tag{72}$$

where $P_f = P(\xi_f)$ is the self-similar pressure at the heat front.

This determines the piston shock parameters:

$$\boxed{p_{0,s} = P_f \, A^{a_3} \, B^{b_3}, \qquad \tau_s = c_3.} \tag{73}$$

That is, the pressure boundary condition for the piston shock problem is directly inherited from the subsonic heat wave solution.

7.3 Position Patching

In the subsonic ablation problem alone, the ablation front sits at $x_f(t) \equiv 0$. In the coupled problem, the ablation front instead sits at the piston position of the shock solution:

$$x_f(t) \equiv x_{\text{shock}}(t, \, m_f(t)). \tag{74}$$

The complete Eulerian position is therefore:

$$\boxed{x(t,m) = \begin{cases} x_{\text{heat}}(t,m) + x_f(t) & \text{if } m < m_f(t) \quad \text{(ablated region)}, \\ x_{\text{shock}}(t,m) & \text{if } m \geq m_f(t) \quad \text{(shocked/unshocked region)}, \end{cases}} \tag{75}$$

where:

- $x_{\text{heat}}(t,m)$ is the position from the *pure* subsonic ablation solution (Eq. (36)), which by itself would place the ablation front at $x = 0$.
- $x_f(t) = x_{\text{shock}}(t, m_f(t))$ is the position of the ablation front (the "piston" of the shock), computed from the shock solution.
- $x_{\text{shock}}(t,m)$ is the position from the piston shock solution, valid from the piston ($m = m_f$) through the shock front and into the unperturbed region.

7.4 Hydrodynamic Profile Patching

For the hydrodynamic profiles (density, velocity, pressure, temperature), the patching is simpler since the profiles are continuous (though not necessarily smooth) across the ablation front:

$$\boxed{h(m,t) = \begin{cases} h_{\text{heat}}(m,t) & \text{if } m < m_f(t), \\ h_{\text{shock}}(m,t) & \text{if } m \geq m_f(t), \end{cases}} \tag{76}$$

where h denotes any of the hydrodynamic fields: v, u, p, or derived quantities such as $T = (pv^{1-\mu}/(rf))^{1/\beta}$.

7.5 Summary of the Patching Procedure

The full subsonic heat wave + shock solution is constructed as follows:

1. **Solve the subsonic ablation problem**: Find ξ_f, P_f, and the self-similar profiles $V(\xi)$, $U(\xi)$, $P(\xi)$ by the double-shooting method (find P_f such that $P(0) = 0$, then find ξ_f such that $T(0) = 1$).

2. **Extract shock parameters**: From the subsonic solution, compute

$$p_{0,s} = P_f\, A^{a_3}\, B^{b_3}, \qquad \tau_s = c_3. \tag{77}$$

3. **Solve the piston shock problem**: Using $p_{0,s}$, τ_s, the initial density ρ_0 (from the unperturbed medium), and $\omega = 0$ (uniform initial density), solve for the shock self-similar coordinate ξ_s and profiles.

4. **Compute the ablation front position**: Using the shock solution, evaluate $x_f(t) = x_{\text{shock}}(t, m_f(t))$.

5. **Assemble the full solution**: Use Eqs. (75) and (76) to construct the complete spatial profiles for any given time t.

Key physical insight

The subsonic heat wave problem provides the pressure history at the ablation front, which acts as the "piston" that drives the shock. The ablation front mass $m_f(t)$ defines the interface between the two regions. In the sub region ($m < m_f$), radiation diffusion dominates and the material is expanding (being ablated). In the shock region ($m > m_f$), the dynamics is purely hydrodynamic (adiabatic compression by the piston).

8 Asymptotic Solution Near the Ablation Front

For completeness, the asymptotic behaviour near $\xi \to \xi_f$ is:

$$V(\xi) \approx V_*\, (\xi_f - \xi)^{1/q}, \tag{78}$$

$$U(\xi) \approx c\, \xi_f\, V(\xi), \tag{79}$$

$$P(\xi) \approx P_f - c^2\, \xi_f^2\, V(\xi), \tag{80}$$

with:

$$V_* = \left[-\left(1+\frac{1}{r}\right) \frac{q\,c}{k}\, \xi_f\, P_f^{1-n} \right]^{1/q}. \tag{81}$$

These provide the initial conditions for the numerical ODE integration from ξ_f inward to the origin.

9 Explicit Position Formulas for the Two Test Cases

We specialize the general formulas to the two verification test cases: both use **Gold (Au)** with identical material parameters but different temporal drive exponents τ:

Parameter	Fig 8	Fig 9
τ	0	0.123
α	1.5	1.5
β	1.6	1.6
λ	0.2	0.2
μ	0.14	0.14
γ	1.25	1.25
ω	0	0
ρ_0	19.32 g/cm^3	19.32 g/cm^3
T_0	1 HeV	1 HeV

All formulas below assume $\omega = 0$ (uniform initial density).

9.1 Dimensional Constants

$$A = T_0\,(r\,f)^{1/\beta}, \qquad B = \frac{16\,\sigma_{\rm sb}\,g}{3\,\beta\,(r\,f)^n}, \qquad v_0 = \frac{1}{\rho_0}, \qquad p_{0,s} = P_f\, A^{a_3}\, B^{b_3}. \tag{82}$$

9.2 The Five Position Functions

9.2.1 1. Boundary Position: $x_{\rm boundary}(t)$

The deepest point reached by the ablation heat wave (where $m = 0$):

$$\boxed{x_{\rm boundary}(t) = \frac{A^{a_2}\, B^{b_2}}{c_2+1}\, U(0)\;\, t^{\,c_2+1}} \tag{83}$$

Since $U(0) < 0$, the boundary moves in the negative x direction.

9.2.2 2. Subsonic Region Position: $x_{\text{sub}}(m,t)$

Position of a Lagrangian element with mass $m < m_f(t)$ inside the ablated region:

$$\boxed{x_{\text{sub}}(m,t) = x_{\text{ablation}}(t) + \frac{A^{a_2}\,B^{b_2}}{c_2+1}\Big[U\big(\xi(m,t)\big) - c\,\xi(m,t)\,V\big(\xi(m,t)\big)\Big]\;t^{\,c_2+1}} \tag{84}$$

where $\xi(m,t) = m\,A^a\,B^b\,t^c$ is the subsonic self-similar coordinate, and $U(\xi), V(\xi)$ are the subsonic self-similar velocity and specific volume profiles.

9.2.3 3. Ablation Front Position: $x_{\text{ablation}}(t)$

The ablation front $m = m_f(t)$ separates the ablated and shocked regions. Its Eulerian position is obtained by evaluating the *shock* solution at the ablated mass:

$$\boxed{x_{\text{ablation}}(t) = (v_0\,p_{0,s})^{1/2}\;t^{\,(2+\tau_s)/2}\left[\hat{\xi}(t)\,V_s\big(\hat{\xi}(t)\big) + \frac{2}{\tau_s+2}\,U_s\big(\hat{\xi}(t)\big)\right]} \tag{85}$$

where $V_s(\hat{\xi})$, $U_s(\hat{\xi})$ are the *shock* self-similar profiles, and $\hat{\xi}(t)$ is the self-similar coordinate of the ablated mass in the shock frame:

$$\hat{\xi}(t) = m_f(t)\left(\frac{v_0}{p_{0,s}}\right)^{1/2} t^{-(2+\tau_s)/2} = \underbrace{\xi_f\,A^{-a}\,B^{-b}\left(\frac{v_0}{p_{0,s}}\right)^{1/2}}_{C_{\hat{\xi}}}\;t^{\overbrace{-c-(2+\tau_s)/2}^{d_{\hat{\xi}}}}. \tag{86}$$

Key point

The exponent $d_{\hat{\xi}} = -c-(2+\tau_s)/2 \neq 0$ in general. This means the ablated mass's coordinate in the shock frame is **time-dependent**: the combined sub+shock problem is *not* fully self-similar. Consequently, $x_{\text{ablation}}(t)$ is *not* a simple power law $C\,t^d$, but a composite expression involving the shock profiles evaluated at a time-varying argument.

9.2.4 4. Shock Region Position: $x_{\text{shock}}(m, t)$

Position of a Lagrangian element $m > m_f(t)$ in the shocked region (for $\omega = 0$):

$$\boxed{x_{\text{shock}}(m,t) = (v_0\, p_{0,s})^{1/2}\ t^{(2+\tau_s)/2} \left[\xi_s(m,t)\, V_s\big(\xi_s(m,t)\big) + \frac{2}{\tau_s + 2}\, U_s\big(\xi_s(m,t)\big)\right]} \tag{87}$$

where $\xi_s(m,t) = m \left(\frac{v_0}{p_{0,s}\, t^{\tau_s+2}}\right)^{1/2}$ is the shock self-similar coordinate.

For mass elements in the unshocked region ($\xi_s > \xi_s^{\text{front}}$): $x(m,t) = v_0\, m$ (initial position).

9.2.5 5. Shock Front Position: $x_{\text{shock front}}(t)$

The outer edge of the shock wave (for $\omega = 0$):

$$\boxed{x_{\text{shock front}}(t) = (v_0\, p_{0,s})^{1/2}\ \xi_s\ t^{(2+\tau_s)/2}} \tag{88}$$

where ξ_s is the (constant) self-similar shock coordinate.

9.2.6 Ablated Mass

$$m_f(t) = \xi_f\, A^{-a}\, B^{-b}\ t^{-c} \tag{89}$$

9.3 Numeric Coefficients for the Two Test Cases

Let each power-law position be written as $x_i(t) = C_i\, t^{d_i}$ where applicable.

9.3.1 Fig 8: $\tau = 0$ (Constant Boundary Temperature)

Quantity	Coefficient C_i	Time power d_i
x_{boundary}	-6.411×10^7	1.0365
$x_{\text{shock front}}$	6.060×10^3	0.7760
m_f	4.454×10^1	0.5156

Self-similar constants: $\xi_f = 0.4277$, $P_f = 0.3871$, $U(0) = -10.544$.
Shock parameters: $p_{0,s} = 2.540 \times 10^8$ Ba, $\tau_s = -0.4479$, $\xi_s = 1.671$.
Ablation front coordinate: $C_{\hat{\xi}} = 6.389 \times 10^{-1}$, $d_{\hat{\xi}} = -0.2604$.

9.3.2 Fig 9: $\tau = 0.123$ (Constant Flux Temperature)

Quantity	Coefficient C_i	Time power d_i
x_{boundary}	-2.697×10^8	1.1245
$x_{\text{shock front}}$	1.293×10^5	0.9377
m_f	4.787×10^3	0.7510

Self-similar constants: $\xi_f = 0.3502$, $P_f = 0.4372$, $U(0) = -7.760$.
Shock parameters: $p_{0,s} = 2.335 \times 10^{11}$ Ba, $\tau_s = -0.1245$, $\xi_s = 1.177$.
Ablation front coordinate: $C_{\hat{\xi}} = 7.142 \times 10^{-2}$, $d_{\hat{\xi}} = -0.1868$.

9.4 Summary of All Exponent Relations

For convenient reference, the temporal exponents of the five position functions are given by the following expressions (valid for $\omega = 0$):

Position	Time exponent d	Expression
x_{boundary}	$c_2 + 1$	$\frac{3\mu-2\lambda-2+\tau[2(\beta-\alpha-4)-\beta\lambda+\mu(4+\alpha)]}{4+2\lambda-4\mu} + \mu$
$x_{\text{shock front}}$	$\frac{2+\tau_s}{2} = \frac{2+c_3}{2}$	(from shock)
m_f	$-c$	$\frac{2+2\lambda-3\mu-\tau[2(\beta-\alpha-4)-\beta\lambda+\mu(4+\alpha)]}{4+2\lambda-4\mu}$
x_{ablation}	composite	not a single power law